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Lab 10.3 Machine Learning 3 - Real-time Sound Recognition for Classification

Entire remote monitoring system

3.1 MTConnect for pump monitoring

In the prvious section, we performed utilizing the CNN model by TensorFlow Lite to predict the running state of the vacuum pump. Let's make an entire monitoring system as Figure 1. Depending on the project or system requirements, you skip the MTConnect part. However, considering the potential use of MTConnect to collect machine data in the future, design and create an MTConnect agent to collect and merge the running state prediction (Event) and sound level (Sample) of the vacuum pump.

The idea is that we know both the execution state (OFF/ON, Event) and the vacuum state (Vacuuming/Air leaking, Event) from the CNN model. Also, we calculate the sound level (Sample). The schematic and data items are shown in Figure 7.

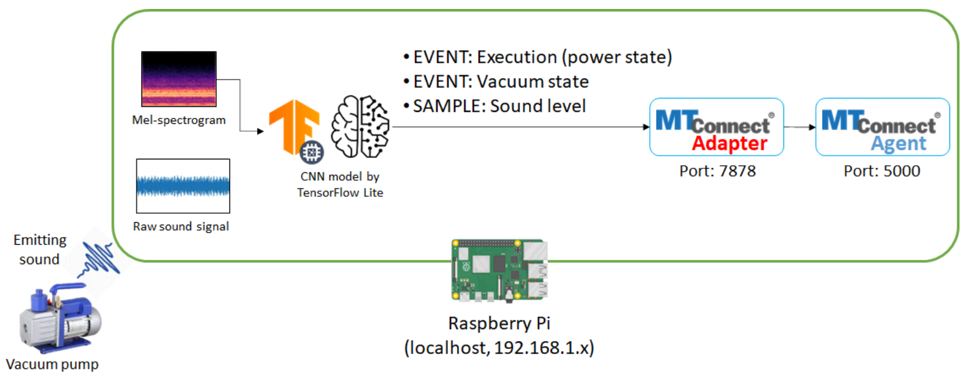


Figure 7 Schematic and data items of MTConnect

The examples of 'agent.cfg' and 'device.xml' are below. You may have different an XML data structure or data item definitions. Also, you can check the example of MTConnect agent running on the server computer: http://mepotrb16.ecn.purdue.edu:5000/. Recall the previous labs to run MTConnect service on Raspberry Pi and reuse the agent execution file and other directory/files (schema, style, and so on).

---

agent.cfg

```
jsonc
Devices = device.xml
AllowPut = true
Port = 5000
ReconnectInterval = 1000
BufferSize = 17
SchemaVersion = 1.4
MonitorConfigFiles = true
Pretty = true
# MinimumConfigReloadAge = 30

Adapters {
   # Log file has all machines with device name prefixed
```

```
        Device {
            Device = vacuum_pump
            Host = 127.0.0.1
            Port = 7878
        }
    }

    Files {
        schemas {
            Path = ./schemas
            Location = /schemas/
        }
        styles {
            Path = ./styles
            Location = /styles/
        }
        Favicon {
            Path = ./styles/favicon.ico
            Location = /favicon.ico
        }
    }

    DevicesStyle { Location = /styles/Devices.xsl }
    StreamsStyle { Location = /styles/Streams.xsl }

    # Logger Configuration
    logger_config
    {
        logging_level = debug
    }
```

---

device.xml

```xml
<?xml version="1.0" encoding="UTF-8"?>
<MTConnectDevices xmlns:m="urn:mtconnect.org:MTConnectDevices:1.1"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns="urn:mtconnect.org:MTConnectDevices:1.1"
xsi:schemaLocation="urn:mtconnect.org:MTConnectDevices:1.1
http://www.mtconnect.org/schemas/MTConnectDevices_1.1.xsd">
        <Header creationTime="2010-03-04T18:44:40+00:00" sender="localhost"
instanceId="1267728234" bufferSize="131072" version="1.1"/>
        <Devices>
                <Device id="vacuum_pump" name="vacuum_pump" uuid="001">
                        <DataItems>
                                <DataItem category="EVENT" id="001" name="avail"
type="AVAILABILITY"/>
                                <DataItem category="EVENT" id="e1" name="execution"
type="EXECUTION"/>
                                <DataItem category="EVENT" id="vs1" name="vacuum_state"
type="STATE"/>
                                <DataItem category="SAMPLE" id="spl" name="sound_level"
type="SOUND_LEVEL" units="DECIBEL"/>

                        </DataItems>
                </Device>
        </Devices>
</MTConnectDevices>
```

The MTConnect adapter sample code (lab10_sample_adapter.py) is also given. The interpretation algorithm to take the CNN model output as MTConnect data item is shown in Figure 8. If the output is 0, execution is "OFF" and vacuum state is

"N/A" (not available). If the output is 1, execution is "ON" and vacuum state "Air leaking". Lastly, if the output is 2, execution is "ON" and vacuum state "Vacuuming". However, it is recommended to try programming the adapter yourself. If you use the given sample code, you have to modify some lines. The incomplete part of the sample script is below. You have to change the variables which is wrarpped by aserteric marks (*). By completing the sample code , perform Task 3.1.

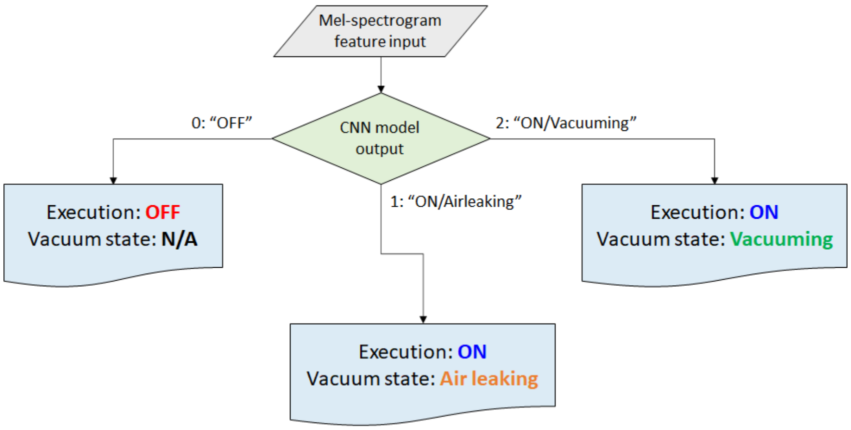


Figure 8 Algorithm to take MTConnect data item from CNN model output

Note that your CNN model (h5 format), mtconnect_adapter.py, and data_item.py must be in the same directory with the adapter.

Python - Python3 (Part of lab10_sample_adapter.py)

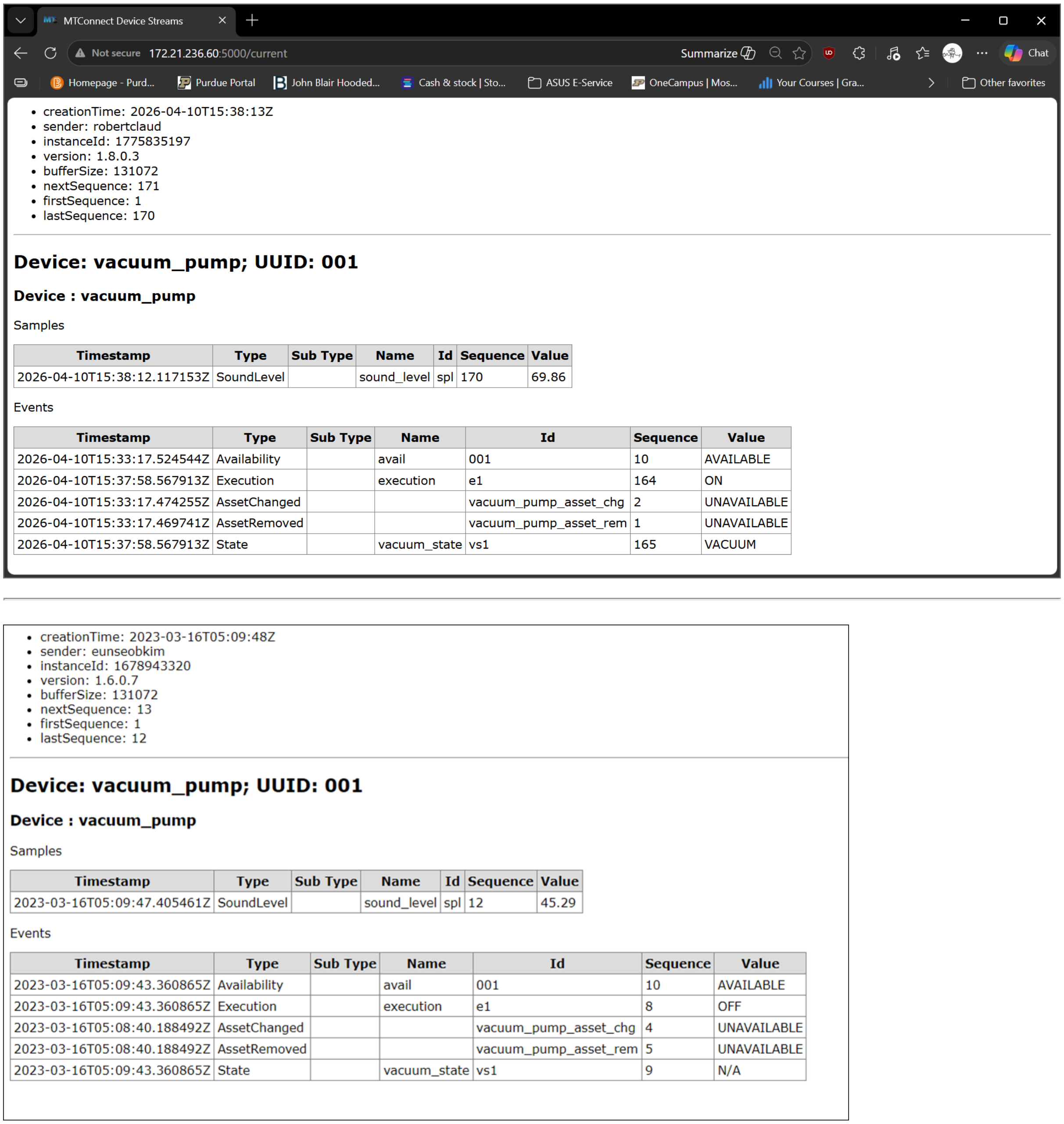
```
## == GLOBAL CONSTANT ==
SAMPLE = "sample" # sample string for MTConnect HTML sample method
CURRENT = "current" # current string for MTConnect HTML current method
SAMP_RATE = int(??) # sampling rate
*SAMP_RATE* = int(?) # sampling rate
*CHUNK* = int(?) # chunk size
*AGENT* = "http://?ip?:?port?/" # MTConnect agent ip:port

*N* = int(?) # number of sequence to take sound
*N_FFT* = int(?) # number of FFT
*N_MELS* = int(?) # number of Mel filter bank

*model_file* = '?20230309_212154_Prelab10_CNN_model1.h5?' # CNN model fileanme. Note that
this file must be in the same directory
```

Task 3.1

Run MTConnect agent and adapter on Raspberry Pi with given informationa and sample. Capture '/current' response of the MTConnect agent as Figure 9 from **your laptop** and attach it to the report. While performing this, check if your adapter is working correctly. You see the terminal as Figure 10. **If nothing is shown in your browser, please close the "Pretty" style following the instruction of Task 2.1 in L4 Colab 2.**



MTConnect Device Streams

Not secure 172.21.236.60:5000/current

- creationTime: 2026-04-10T15:38:13Z
- sender: robertclaud
- instanceId: 1775835197
- version: 1.8.0.3
- bufferSize: 131072
- nextSequence: 171
- firstSequence: 1
- lastSequence: 170

Device: vacuum_pump; UUID: 001

Device : vacuum_pump

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-04-10T15:38:12.117153Z	SoundLevel		sound_level	spl	170	69.86

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-04-10T15:33:17.524544Z	Availability		avail	001	10	AVAILABLE
2026-04-10T15:37:58.567913Z	Execution		execution	e1	164	ON
2026-04-10T15:33:17.474255Z	AssetChanged			vacuum_pump_asset_chg	2	UNAVAILABLE
2026-04-10T15:33:17.469741Z	AssetRemoved			vacuum_pump_asset_rem	1	UNAVAILABLE
2026-04-10T15:37:58.567913Z	State		vacuum_state	vs1	165	VACUUM

- creationTime: 2023-03-16T05:09:48Z
- sender: eunseobkim
- instanceId: 1678943320
- version: 1.6.0.7
- bufferSize: 131072
- nextSequence: 13
- firstSequence: 1
- lastSequence: 12

Device: vacuum_pump; UUID: 001

Device : vacuum_pump

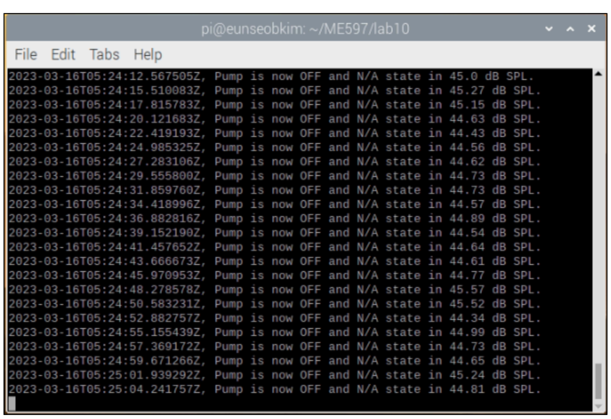
Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2023-03-16T05:09:47.405461Z	SoundLevel		sound_level	spl	12	45.29

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2023-03-16T05:09:43.360865Z	Availability		avail	001	10	AVAILABLE
2023-03-16T05:09:43.360865Z	Execution		execution	e1	8	OFF
2023-03-16T05:08:40.188492Z	AssetChanged			vacuum_pump_asset_chg	4	UNAVAILABLE
2023-03-16T05:08:40.188492Z	AssetRemoved			vacuum_pump_asset_rem	5	UNAVAILABLE
2023-03-16T05:09:43.360865Z	State		vacuum_state	vs1	9	N/A

Figure 9 MTConnect agent '/current' response



pi@eunseobkim: ~/ME597/lab10

File Edit Tabs Help

```
2023-03-16T05:24:12.567505Z, Pump is now OFF and N/A state in 45.0 dB SPL.
2023-03-16T05:24:15.510083Z, Pump is now OFF and N/A state in 45.27 dB SPL.
2023-03-16T05:24:17.815783Z, Pump is now OFF and N/A state in 45.15 dB SPL.
2023-03-16T05:24:20.121683Z, Pump is now OFF and N/A state in 44.63 dB SPL.
2023-03-16T05:24:22.419193Z, Pump is now OFF and N/A state in 44.43 dB SPL.
2023-03-16T05:24:24.985325Z, Pump is now OFF and N/A state in 44.56 dB SPL.
2023-03-16T05:24:27.283106Z, Pump is now OFF and N/A state in 44.62 dB SPL.
2023-03-16T05:24:29.555800Z, Pump is now OFF and N/A state in 44.73 dB SPL.
2023-03-16T05:24:31.859760Z, Pump is now OFF and N/A state in 44.73 dB SPL.
2023-03-16T05:24:34.418996Z, Pump is now OFF and N/A state in 44.57 dB SPL.
2023-03-16T05:24:36.882816Z, Pump is now OFF and N/A state in 44.89 dB SPL.
2023-03-16T05:24:39.152190Z, Pump is now OFF and N/A state in 44.54 dB SPL.
2023-03-16T05:24:41.457652Z, Pump is now OFF and N/A state in 44.64 dB SPL.
2023-03-16T05:24:43.666673Z, Pump is now OFF and N/A state in 44.61 dB SPL.
2023-03-16T05:24:45.970953Z, Pump is now OFF and N/A state in 44.77 dB SPL.
2023-03-16T05:24:48.278578Z, Pump is now OFF and N/A state in 45.57 dB SPL.
2023-03-16T05:24:50.583231Z, Pump is now OFF and N/A state in 45.52 dB SPL.
2023-03-16T05:24:52.882757Z, Pump is now OFF and N/A state in 44.34 dB SPL.
2023-03-16T05:24:55.155439Z, Pump is now OFF and N/A state in 44.99 dB SPL.
2023-03-16T05:24:57.369172Z, Pump is now OFF and N/A state in 44.73 dB SPL.
2023-03-16T05:24:59.671266Z, Pump is now OFF and N/A state in 44.65 dB SPL.
2023-03-16T05:25:01.939292Z, Pump is now OFF and N/A state in 45.24 dB SPL.
2023-03-16T05:25:04.241757Z, Pump is now OFF and N/A state in 44.81 dB SPL.
```

Figure 10 Capture of terminal in MTConnect adapter is running

3.2 MySQL database and Grafana dashboard

Let's move on to the data aggregation for MySQL database and visualization by Grafana. To collect MTConnect agent data, use the sample Python script (mtconnect_collecor.py).

MySQL and Grafana information is the same as the previous lab. However they are summarized again below.

- **MySQL**
    - DNS: mepotrb16.ecn.purdue.edu
    - Port: 3306
    - Account: yourname (e.g., johndoe)
    - Database (schema): ME597Fall24
    - Table: yourname_lab6 (e.g., johndoe_lab6)
    - Reuse the table you created in Lab 6.
- **Grafana**
    - DNS: mepotrb16.ecn.purdue.edu
    - Port: 3000
    - Grafana web page URL: http://mepotrb16.ecn.purdue.edu:3000/
    - Account (username): yourname (e.g., johndoe) OR your purdue email account

The sample Grafana dashboard is 'Lab10 sample' in '0_SAMPLE' dashboard folder as Figure 11. Perform Task 3.2 to conclude this lab.

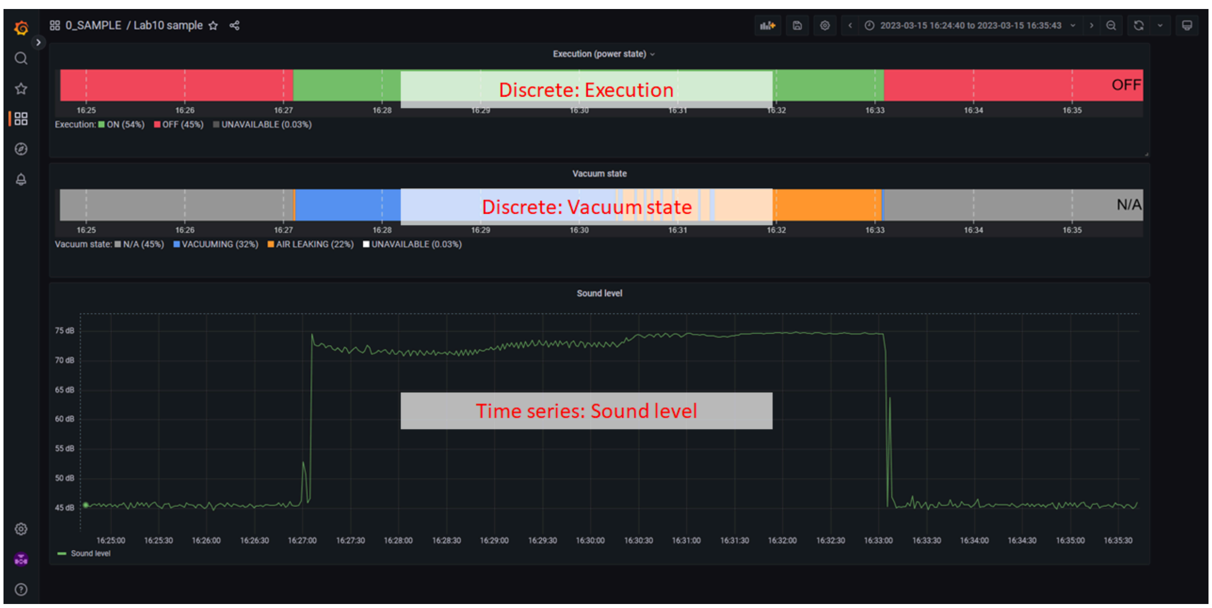


Figure 11 Lab10 Grafana dashboard sample

Task 3.2

Create a dashboard as Figure 11 in your folder created in Lab 7 and save it as the dashboard name of 'lab10 your name', e.g., 'lab10 john doe'. The requirements of the dashboard are below.

- Discrete panel for Execution
- Discrete panel for Vacuum state
- Time series panel for Sound level
- (Optional) Add more if you want

Capture the dashboard and attach it to the report.

After creating the dashboard, confirm if the entire monitoring system works in real time by refreshing the dashboard. TA will change the operational state of the pump during lab occasionally.

Task 3.3 Further discussion

Probably, sometimes you may see false results as Figure 12. In this example, although all actual state (red circle in Figure 12) was Air leaking, sporadically the inference state might be Vacuuming or vice versa. Discuss how to fix this problem in any aspect. For instance, we can discuss various aspects such as machine learning/deep learning architecture selection, training, algorithm, and so on.

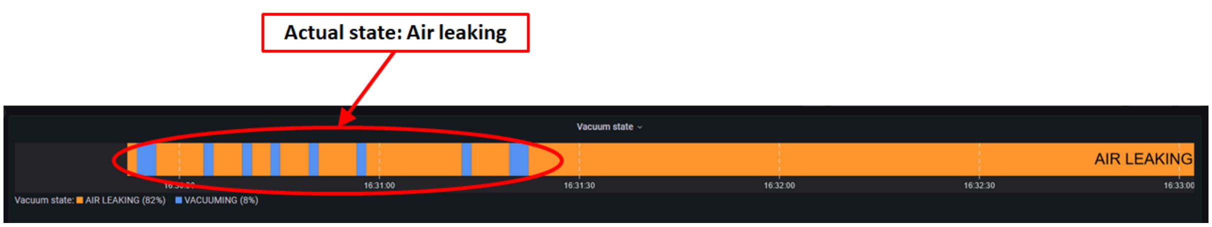


Figure 12 False prediction result

This issue stems from the convolutional nerual network predicting the wrong outcome. Essentially, the fix would be to improve the performance of the CNN. Therefore, a few fixes could be implemented:

1. Increase the amount of training data. This would potentially capture similar conditions where the current model predicts the incorrect outcome.
2. Increase the number of training epochs. This would adjust the model to better predict the outcome of the data. However, the number should not be increased too far to avoid overfitting.
3. Increase the window size of sampling. This would allow each sequence of training to capture more of the sound pattern.

Lab10 Summary and Deliverables

Answer the following questions for your achievements

Q1. Please summarize Lab10.

In Lab 10 we learned how to train and use a convolutional neural network to interpret sound readings using a mel-spectogram. The sound it was trained on was the sound coming off of a vaccum pump which has the three distinct possible states:

- OFF
- ON/VACUUM
- ON/AIR LEAKING We then deployed this CNN to an edge device (Raspberry Pi) to test out tensorflow lite, to see how it could be used to process data with far fewer computing resources. Using the raw sound level output, and the predictions of the data into three different states, we used the edge device to parse the data into a MySQL database, and then built a dashboard to allow live updates of the state and sound level of the vacuum pump.

Q2. What skills did you have to develop to accomplish this project?

To accomplish this project, I needed to develop my skills in training and understanding how machine learning models work. Specifically, what hyperparameters to adjust when training these models and how to deploy them to edge devices.

Q3. What aspects of this project were the most beneficial for your learning?

Sending the data to a Grafana dashboard was the most beneficial for my learning since this allowed me to see exactly how the machine learning predicted the state compared to the graphical sound level. Also, seeing the graphs of the MEL-Spectogram helped since I could train my own eyes on the same data as the model.

Q4. What challenges did you encounter in completing the project?

I encountered some challenges using tensorflow, specifically with the default version of flatbuffers.

Q5. How did you overcome the challenges or remedy the problems encountered?

First, I consulted Google Gemini for potential known fixes to incompatibilities with TensorFlow Lite and Flatbuffers. It suggested I needed to upgrade my version of Flatbuffers to 2.0. This fixed one issue but introduced a new issue where imp was not a package included with python as of version 3.12.0. Since I am using Python 3.13.5, I needed to modify the Flatbuffer installation to no longer need imp. I simply changed the file compat.py in flatbuffers to use importlib.util instead of imp.

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